THE WHITE ELEPHANT

Alan and Eric run the White Elephant hotel.

This was bought at a low price because it was quite run down and did not have very high occupancy rates. Alan and Eric aim to increase the fortunes of the hotel and have recently completed a refurbishment programme to improve the property. Keeping all rooms in top condition is of upmost importance. They are now targeting a 90% occupancy rate although at the moment there is no definite idea of the number of bookings being made. Eric feels that at least 35% of all bookings come from repeat bookings.

Alan is keen to increase the number of bookings from local companies. At the moment there is no provision for corporate guests but Alan feels 40% of all bookings should be for corporate guests.

There are 23 rooms. At the moment the hotel has records on about 500 guests who have stayed in the hotel but this is expected to increase to 1,500. Rooms vary according to the number of beds (single, twin or double) the facilities (bath, shower, toilet) and the size (small, normal, large).

Eric is responsible for day-to-day hotel operations. Alan looks after the financial side of things as well as looking after corporate clients.

The hotel has a basic computerised room booking system used by the receptionists when potential guests ring the hotel to book a room. This system was inherited from the previous owners of the hotel.Neither the system itself nor the current booking procedures are documented. Alan and Eric want to replace this system with a new one that includes a website allowingguests to book rooms online. They feel this is necessary to support the drive for increased occupancy by improving the booking process including provision for online bookings and bookings from corporate guests and the provision of quality rooms.

Eric had a bad experience with IT from a previous company where the large amount of documentation caused confusion amongst everybody concerned and the provider failed to deliver a number ofagreed requirements.

Alan and Eric have met the local IT company that created the current system and were impressed with their technical expertise and business knowledge. They have decided to seek their advice and have asked them to quote for design and development of the system.

They recommended a business analyst to perform an initial investigation and their findings are detailed below. Your role is to continue with the project with further analysis of these requirements. The original business analyst has provided a number of templates for use on the project and is on hand to provide further support if needed.

Eric wants guests to be able to view, select, and book rooms. All booking should be guaranteed by a credit card. There are six receptionists who will also use the system to make bookings for clients who would rather ring the hotel than use a web-based system.Receptionists will follow the same process as guests but enter the information on their behalf. The reception staff members havelittle computer experience and Eric is not sure how they will cope. Eric would therefore like the system to be easy to use.

In order to determine the success of the web-based system,Eric wants to be able to view a report showing the number of successful bookings made each month by guests or receptionists.

Eric feels that speed is an essential component and that a slow system will put people off. He feels that accommodation searches should be completed within 30 seconds.

Alan thinks it is important that the system can handle bookings from corporate guests. Corporate guests are businesses who want to be able to book rooms for employees and pay for them later. Alan is also aware that the current system only allows guests to book one room at a time. This is OK when being discussed over the phone with the receptionist but when using the web a corporate client must be able to book more than one room. Alan wants accounts to invoice corporate guests on a monthly basis for bookings made during the month. He is also worried about hackers getting into the system.

Alan would like the new system to be able to produce an ad-hoc report on the number of rooms booked and the type of guest (private or business). He would like to be able to select the start and end date of the report. No other member of staff should be able to access this report.

The six reception staff members work one of three shifts, 6:00 to 14:00, 8:00 to 18:00 or 14:00 to 22:00. Reception closes at 22:00. Eric who lives in a private flat in the hotel is always on call to handle out of hours queries.

The maintenance engineer works from 9:00 to 17:00 is responsible for fixing all faults reported by the cleaner and any other maintenance work that is required.

Eric has noticed that some maintenance issues are going unresolved and would like the new system to be able to record maintenance issues. He feels that the details of each fault on the fault recording sheet should be entered into the system and then a report on all unresolved issues can be printed by the hotel

manager and given to the maintenance engineer. Eric would also like a report that shows the average time taken to repair a fault.

Eric employs a hotel manager, who is very experienced and has worked in the hotel and catering industry for twenty three years, to look after the day to day running of the hotel. This includes all necessary administration and supervision of the receptionists, cleaners, maintenance engineer, and chef.Because of the nature of his position the hotel manager holds a lot of confidential information about the staff at The White Elephant. The hotel manager wants to be able to print two reports at the end of each day. The first: listing all guests who are checking out in the morning to allow the preparation of the room for the next guest. The second; to print all guests who will be taking breakfast in the morning so that the chef knows how many breakfasts need to be prepared.

The hotel also employs two cleaners, one maintenance engineer, and a chef. The cleaners start work at 6:00 cleaning the restaurant and rooms and reporting any maintenance issues to the maintenance engineer. The cleaners currently record faults on a fault recording sheet which is passed to either the hotel manager or Eric, depending on who is on duty when cleaning has finished to be passed to the maintenance engineer.

The hotel manager will either work from 6:00 to 15:00 or 10:00 to 20:00 and will support the reception staff when they are busy. Eric tends to work the opposite shift to the hotel manager to ensure a responsible manager is always present from 6:00 to 20:00.

All of the existing staff are extremely busy and need to work regular overtime. Nobody has any relevant IT experience.

Alan does not live at the hotel and spends a lot of time working from home about twenty miles away. He spends a lot of time reviewing finances and making financial projections as well as canvassing corporate guests and discussing marketing possibilities with local marketing companies. He would like the new system to be able to record company details so that these are already on the system when corporate guests make a booking.

Eric has suggested the required process forbooking a room will be as follows:

Start of required booking process.

When a guest wants to book a room, the system will display the room booking screen. Using this screen, the user will enter the dates required and the type of room (single, twin, or double). The system will then

check for available rooms. If no rooms are available the system will give the guest the opportunity to enter a different room type.

If rooms are available the system will list all the available rooms, their facilities, and the price for the room. The guest will choose one of these rooms and then enter their personal details when prompted. If they are an existing guest they will be able to retrieve their personal details. Personal details include their name, address details, telephone number, and email address.

The system will then use a payment gateway to request the credit card details of the client and attempt to authorise the card.

If authorisation is refused the system will display an “authorisation failed” message and give the option to use another credit card.

If authorisation is successful, the system will display a unique booking reference and also send a confirmation email to the guest.

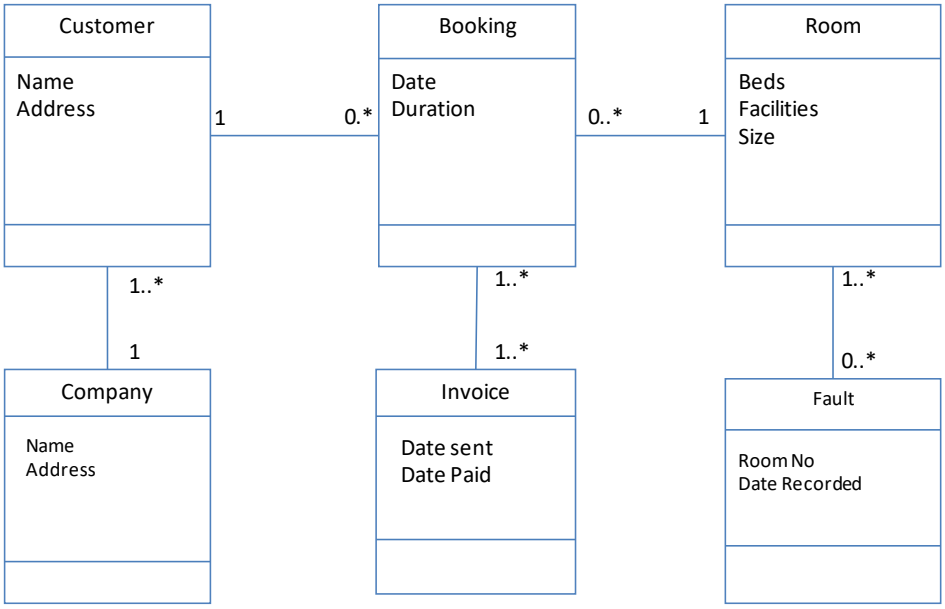
End of required booking process.

Initial list of requirements

From this information an initial list of requirements has been produced.

1. Guests will be able to view available rooms.
2. Guests will be able to select and book a room.
3. The system shall use Microsoft Outlook to send an email to confirm a room has been booked.
4. The system shall be able to confirm a room has been booked within thirty seconds.
5. The system shall take payment by credit card for all bookings.
6. Directors will be able to view the number of successful bookings made each month.
7. Corporate guests shall be able to book more than one room.
8. Alan shall be able to invoice corporate guests at the end of each month.
9. The system shall be easy to use.
10. Reception staff shall be able to use the system to view bookings.
11. The hotel manager shall be able to print a list of guests checking out the following morning.
12. The hotel manager shall be able to print a list of guests who require breakfast the next morning.
13. The hotel manager shall be able to enter details of room faults.
14. The hotel manager shall be able to print a list of faults that need to be resolved.
15. Directors shall be able to produce a report that shows the average time taken to repair faults.
16. The system shall be able to capture guest details.
17. The system shall be able to capture company details.

Class Diagram
WE Hotel booking System



Exercise One:

a) Identify **three** stakeholders that you would wish to involve in the project.

b) State **one** piece of information concerning the requirements that you would wish to elicit from **each of the three** stakeholders

c) Suggest and justify **one** suitable technique you would use to elicit this information

Ans:

The three stakeholders I would include in this are :

1) **Eric (Operation Manager): The information that I can elicit from him is the way receptionists handle bookings and where they struggle**
   Structured - Interview technique can be used cause Eric understand daily operations, system problems and performance concerns. This ensures accurate requirements and avoids misunderstandings
2) **Alan(Director – Finance & Corporate Clients) : He wants more customers to be from Corporate so we can ask him on what corporate invoices we should work(billing, rooms, etc)**
   Structured – Interview technique can be used to get what exactly the Finance planning is and what exactly he want us to do to get Corporate Clients
3) **Receptionists Representative: We would know what actually ‘ easy to use’ means from their perspectives**
   Observation technique can be used to observe what are the previous techniques they are using, what they already know and what knowledge we need to transfer. So that system becomes easy to use

Exercise Two

Refer to the initial list of requirements and

a) Identify **two** functional requirements.

b) Identify **two** non-functional requirements. For **each one,** explain why it is non-functional. – Give type of NFR

c) Identify **two** requirements that are potentially in conflict with each other and explain why they are in conflict.

d) Identify **one** example of an ambiguous requirement and explain why it is ambiguous.

e) Identify one solution and state why it is a solution.

Ans:

a) Functional Requirements:
   1) Guests should be able to view available room
   2) Guests will be able to select and book rooms
b) Non – Functional Requirements:
   1) The system shall be able to confirm a room has been booked within thirty seconds

   This requirement is Non – Functional Cause it tells us about the performance of the system and does not tell us to follow an operation. This requirement is Non – Functional Performance Requirement

   2) The system shall be easy to use

   This requirement is Non – Functional Usability Requirement cause this requirement tell us that the system should be easy to operate
c) We have requirements to be fast as well as secure which can create a conflict since by making a system secure it can slow system response time, which conflicts Erics demand for speed
d) The system shall be easy to use

   This is ambiguous can we cannot interprete what exactly they need. System should be easy – for whom? Receptionist? Clients? Guests?
e) We can rewrite this requirement in SMART way - 'Reception staff shall be able to complete a booking in under 3 minutes'

Exercise: Three

Use the requirements catalogue templates to complete requirement catalogue entries for R1 and R2 from the initial list of requirements for The White Elephant.

Author:	Date Captured:	Version:	Status

ID	R1
Name	View Available Rooms
Description	Guests and Receptionists must be able to view a list of available rooms based on dates and room types
Source	Eric
Owner	
Priority	High
Rationale	To support online booking and to get occupancy of the Rooms
Associated NFRs	System should display list of the available rooms within 30 seconds.
Related documents	List of all Rooms that are present and booking catalogue.
Related Requirements	R2,R4,R7

Author:	Date Captured:	Version:	Status

ID	R2
Name	Corporate multiple rooms booking
Description	System should allow corporate clients to book more than one room in a single transaction
Source	Alan
Owner	
Priority	High
Rationale	Supports corporate business, increases revenue and supports goal of 40% booking should be from corporate
Associated NFRs	We are giving accessibility to the corporate clients to book multiple rooms
Related documents	We can review old booking register to get an idea how corporate clients are booking for what period and number of the rooms.
Related Requirements	R1,R3,R4,R5

Ex**ercise Four**

Consult the class diagram and answer the following questions:

- What information concerning rooms are being recorded?
- Can corporate customers make a booking for more than one room?
- Can the time taken to repair faults be calculated?
- What is wrong with the association between company and customer?
- How many bookings can each invoice cover?
- Comment on the suitability of the attributes for Booking

Ans:

- We are recording beds, facilities given and size of the rooms
- Yes, corporates can book multiple rooms
- No, we do not calculate the time taken to repair faults
- Since it account every customer with a company, or at least 1company with every customer we cant account individual customers.
- Each invoice can cover one or more bookings.
- Current Attributes date and duration can be insufficient, though we should also record booking id, status, cost, etc.

Exercise Five:

When validating requirements, a review group should be defined to confirm the suitability of the requirements prior to sign off.

Identify three stakeholders from the White Elephant who should be a member of this group and suggest what role they would play.

Ans:

1) Eric : Confirms usability, speed, and booking process
2) Alan: Confirms corporate bookings, invoicing , and security
3) Hotel Manager: Confirms maintenance reports and daily operations